

# Factors predicting myopia incidence in China and Europe

Nina Tahhan<sup>1,2</sup>  | Xiangui He<sup>3</sup>  | Kathryn Saunders<sup>4</sup>  | Pelsin Demir<sup>5</sup>  |  
Rebecca Leighton<sup>4</sup>  | Sara McCullough<sup>4</sup>  | Karthikeyan Baskaran<sup>5</sup>  |  
Antonio Filipe Macedo<sup>5</sup>  | Xun Xu<sup>3</sup>  | Padmaja Sankaridurg<sup>2</sup> | Thomas Naduvilath<sup>1,2</sup> 

<sup>1</sup>Brien Holden Vision Institute (BHVI), Sydney, New South Wales, Australia

<sup>2</sup>School of Optometry and Vision Science, University of New South Wales, Sydney, New South Wales, Australia

<sup>3</sup>Shanghai Eye Disease Prevention and Treatment Centre, Shanghai Eye Hospital, Shanghai Vision Health Centre & Shanghai Children Myopia Institute, Shanghai, China

<sup>4</sup>Ulster University, Coleraine, UK

<sup>5</sup>Department of Medicine and Optometry, Linnaeus University, Kalmar, Sweden

## Correspondence

Nina Tahhan, Brien Holden Vision Institute (BHVI), Sydney, New South Wales, Australia. Email: [n.tahhan@bhvi.org](mailto:n.tahhan@bhvi.org)

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## Abstract

**Aims:** To develop a predictive model for myopia incidence using population-based data from Chinese and European children.

**Methods:** Analysis of four longitudinal studies from the UK, Sweden and China was conducted. Data from 4405 non-myopic children aged 6 to 16.8 years with spherical equivalent (SE) refraction from  $-0.49$  D to  $+10.00$  D were analysed. Kaplan–Meier and Cox proportional hazard models were used to evaluate the probability of myopia incidence by 12 and 24 months based on age, sex, parental myopia, cycloplegic SE, axial length and axial length/corneal radius of curvature (AL/CR). Hyperopic reserve was defined as the minimum level of hyperopia required to provide  $<10\%$  probability of developing myopia by 24 months.

**Results:** The cumulative incidence of myopia by the 24th month was 18.8% for Chinese and 6.7% for European ( $p < 0.001$ ) populations. Based on multivariate Cox regression, a greater risk of myopia incidence was seen in Chinese children, younger ages, females, children with parental myopia, less hyperopic SE and higher AL/CR. SE had the highest predictive accuracy (C statistic 0.90). The AL/CR ratio had a greater predictive accuracy (C statistic 0.75) than axial length alone (C statistic 0.63). Predictive accuracy of all variables was similar between ethnicities ( $p > 0.50$ ) apart from axial length, which was higher in Chinese children (C statistic 0.65) versus European children (C statistic 0.55,  $p = 0.04$ ). To avert the risk of myopia, Chinese eyes required a greater hyperopic reserve ranging from 0.5 to 1.5 D compared to European eyes ranging from 0 to 0.5 D, depending on age and other predictive factors.

**Conclusions:** SE is the strongest predictive factor for both ethnicities. The influence of predictive factors is similar between ethnicities/regions though Chinese children have a greater risk of developing myopia and require a higher hyperopic reserve. These data could be useful for developing a predictive tool of myopia incidence.

## KEYWORDS

children, incidence, myopia, predictive, risk factors

## INTRODUCTION

Childhood myopia is a growing global health concern, with its prevalence and related risks rising in many regions,

especially in Asia.<sup>1,2</sup> Myopia often emerges in early childhood and is typically progressive, stabilising only in adulthood.<sup>3</sup> Early onset increases the likelihood of high myopia, which substantially elevates the risk of sight-threatening

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complications, including retinal detachment, glaucoma and myopic maculopathy.<sup>4</sup> Prevention, early detection and proactive management with myopia control therapies are therefore crucial in reducing the global burden of vision loss linked to these complications.<sup>4,5</sup> Among these strategies, delaying the onset of myopia is considered the most impactful, as each additional year without myopia onset is more effective in reducing future progression than multiple years of active management.<sup>6</sup>

Identifying children at higher risk of developing myopia allows for early intervention. This enables the implementation of prevention and treatment strategies, such as increased outdoor time, controlled near work and in some cases, the early initiation of myopia control therapies. The International Myopia Institute (IMI) defines 'pre-myopia' as a state where there is a high likelihood of myopia onset based on a combination of factors such as 'baseline refraction, age and other quantifiable risk factors'.<sup>7</sup> Longitudinal research has demonstrated that children who eventually develop myopia exhibit a more pronounced rate of axial elongation prior to clinical manifestation.<sup>8</sup> These findings support the concept of a 'pre-myopia' stage, in which accelerated eye growth may serve as an early marker of imminent myopia.

There is increasing evidence that myopia control therapies such as low dose Atropine,<sup>9–12</sup> and repeated low level red light therapy (RLRL)<sup>13,14</sup> can reduce the incidence of myopia in pre-myopic children. However, such interventions may have one or more significant drawbacks including cost, inconvenience and side-effects such as light sensitivity, reduced near vision and in the case of RLRL, unknown long term safety.<sup>15</sup> Hence the availability of data and tools with predictive capabilities can facilitate more targeted approaches (i.e., avoiding treating those who may never progress to myopia) and support clinicians with patient awareness, education and informed decision making.

There have been prior studies investigating predictive factors associated with the onset of myopia in pre-myopic children. Some of the predictive factors identified include sex, refraction, axial length, axial length/corneal radius (AL/CR), parental myopia and ethnicity.<sup>16–20</sup> Studies have consistently shown that cycloplegic spherical equivalent (SE) refraction for a given age has the strongest predictive capability over other factors.<sup>16,17,19–22</sup> The addition of other factors does not appear to have a great impact on prediction accuracy as judged by the small changes observed in the area under the receiver operating characteristic (ROC) curve. This has led researchers to provide clinical guideline recommendations in relation to hyperopic reserve thresholds expected for a given age when evaluating a child's risk of myopia onset.<sup>23</sup>

It is clear from the literature that the prevalence of myopia is greatest in Asian populations.<sup>24</sup> The literature also increasingly suggests environmental influences may play a greater role than genetic pre-disposition.<sup>24</sup> Indeed, it has been observed that Asian populations living in Australia have a lower rate of myopia than Asians living in Asia.<sup>25</sup>

### Key points

- For Chinese and European children, the likelihood of myopia onset within 12–24 months can be predicted based on factors including age, sex, parental myopia, baseline refraction and axial length/corneal radius.
- Refraction at baseline was the strongest predictive factor for myopia onset for both Chinese and European children.
- Chinese children have a greater risk of developing myopia and need to have a more hyperopic refraction than European children to avert the risk of myopia onset in the short term.

However, it is unclear how the predictive factors for pre-myopia may differ between different populations. So far, published data on predictive factors for myopia have generally been derived from populations from a specific country. Drawing comparisons between these studies is difficult due to methodological differences, including the definition of myopia used, follow-up period and statistical analyses employed. Whilst Zadnik et al.<sup>16</sup> was a multi-ethnic study and included Asian populations, all data collection was conducted in the USA, and therefore, environmental influences would differ for those residing in Asia. Analysis of data on pre-myopia sourced from more than one country would be a valuable addition to the literature and help inform predictive tools that may be used more broadly in practice.<sup>26</sup>

A recent study assessed whether the Predicting Myopia Onset and Progression (PreMO) risk indicator, developed from a study of white children in Ireland and the UK, could predict myopia in children from Hong Kong and a diverse UK cohort.<sup>27</sup> The calculator demonstrated high sensitivity and specificity for predicting myopia for children living in the UK. However, for the children living in Hong Kong, while the calculator demonstrated high sensitivity, it was only moderately specific in its ability to signal children at risk for future myopia. It would, therefore, be useful to explore predictive data derived directly from Asia and compare that with European data. Therefore, the present study aimed to determine factors that are predictive of myopia onset for Asian children living in China and European children living in Sweden and the UK.

## METHODS

This is a retrospective analysis using population-based longitudinal data from studies conducted in the UK, Sweden and China. Ethics approval for all studies was obtained from institutional human research ethics committees, and the tenets of the Declaration of Helsinki for experimentation

TABLE 1 Details of study protocols.

| Clinical trial/population study           | Study registration identifiers | Cycloplegia protocol                                                                            | Refraction                                               | Ocular biometry               |
|-------------------------------------------|--------------------------------|-------------------------------------------------------------------------------------------------|----------------------------------------------------------|-------------------------------|
| Population Study 1@China                  | 2015KY150                      | 2 drops of cyclopentolate (1%) after corneal anaesthesia, 5 min apart; refraction after 30 mins | Annually using Topcon KR-8900 auto-refractor             | Annually using IOL Master 500 |
| Population Study 2@China                  | NCT02980445                    | 2 drops of cyclopentolate (1%) after corneal anaesthesia, 5 min apart; refraction after 30 min  | Annually using Topcon KR-8900 auto-refractor             | Annually using IOL Master 500 |
| Population Study NICER2@ Northern Ireland | REC/18/0102                    | 1 drop of cyclopentolate (1%) after corneal anaesthesia; refraction after 30 min                | Annually using NVision-k 5001 auto-refractor             | Annually using IOL Master 700 |
| Population Study@Sweden                   | DNR 2018/423–31                | 2 drops of cyclopentolate (1%), 10 min apart; refraction after 30 min                           | Annually using Shin–Nippon NVision-k 5001 auto-refractor | Annually using IOL Master 500 |

on humans were adhered to. Written informed consent was obtained from the parent and/or participant prior to data collection.

Two population-based studies from Shanghai provided data for Chinese children aged 6 to 16 years. Children were followed up for 24 months with two annual visits; the first observational cohort study was conducted from 2013 to 2017 and the second outdoor interventional trial was from 2016 to 2018. Only data from the control arm of the latter study were used for this analysis. Data from European children were from a longitudinal study conducted in Northern Ireland, UK from 2019 to 2021 on children aged 6–7 and 12–13 years, followed up for 12 months and a second European study conducted from 2019 to 2021 on Swedish school children aged 8 to 16 years and followed up for 24 months with two annual visits in Kalmar, Sweden.

Details of study procedures from these studies have been described previously.<sup>28–30</sup> Refraction was measured using auto-refraction under cycloplegia, where a Topcon auto-refractor (KR-8900; Topcon, [Topcon.co.jp](http://Topcon.co.jp)) was utilised for two studies in China and the Shin–Nippon open field auto-refractor (SRW5000 or NVision-K 5001, [rexam.co.jp/eye-care](http://rexam.co.jp/eye-care)) for the European studies. The IOL Master (V3, V5, 500, 700, Carl Zeiss, [zeiss.com](http://zeiss.com)) was used for axial length measurements. Table 1 presents the study identifiers and protocols used for refraction and biometry measurements.

## Statistical methods

Inclusion criteria for this analysis were baseline age range of  $\geq 6.0$  to  $< 17$  years and being non-myopic at baseline. Myopia was defined as SE refractive error of  $\leq -0.50$  D, where SE was computed as sphere + cylinder/2. The threshold of SE  $\leq -0.50$  D used in this study follows the internationally recognised definition of myopia established by the IMI through expert consensus,<sup>7</sup> and is commonly adopted in large epidemiological studies.<sup>31,32</sup> Myopia was defined for an eye, not a participant. Participants who failed to attend the 12-month follow-up visit were excluded from the analysis.

Combined data of the two studies from China were referred to as Chinese data and combined data of the two studies from the UK and Sweden were referred to as European data. Only data for eyes that were non-myopic at baseline were considered. Data from the second study in China was collected for the right eye only, while other studies collected data from both eyes. Data was analysed as eyes.

Myopia incidence over a 24-month period was modelled as time-to-event data using discrete time periods. Myopia incidence was recorded in a binary format, where 1 indicated myopia incidence or 0 for those that remained non-myopic during their follow-up period. The myopia incidence time was recorded as 12 months for those participants who became myopic by the 12-month visit and as 24 months for those participants that were non-myopic up

to the 12-month visit but became myopic by the 24-month visit. The Kaplan–Meier analysis provided the cumulative probability of myopia incidence up to 12 and 24 months, computed as  $1 - \text{cumulative probability of not becoming myopic by 12 and 24 months, respectively}$ . Univariate log-rank Mantel-Cox test compared distributions between factors. Probabilities were plotted graphically for factors that were significant.

Multivariate time-to-event analysis was performed using Cox proportional-hazards regression. The follow-up period of 24 months was divided into two discrete periods of 12 months each. The outcome measure for Cox regression was the expected hazard, which in this analysis was the expected incidence of myopia within a given time interval accounting for the effects of the model factors. The factors analysed included region/ethnicity (Chinese vs. European), age at baseline, sex, parental myopia and ocular measurements at baseline, namely, refractive error, axial length and AL/CR ratio. The model can be expressed as:

$$h(t) = h_0(t) \times \exp(b_1X_1 + b_2X_2 + \dots + b_pX_p),$$

where  $h(t)$  is the expected hazard at time  $t$ ,  $h_0(t)$  is the cumulative baseline hazard computed at mean baseline values of independent factors ( $X_1, X_2, \dots, X_p$ ). The natural logarithm of  $h(t)$  represents a linear combination of baseline hazard and factor combinations. Model coefficients, such as  $b_i$ , represent the change in the expected log of the hazard relative to a unit change in  $X_i$ , holding all other factors constant. The exponent of a factor coefficient was the hazard ratio (relative risk) and probability of myopia and was computed as  $h(t)/(1 + h(t))$ . As discrete time intervals were used in place of continuous time-to-event analysis, the baseline hazard (model intercept) was computed for each discrete 12-month time interval. Additionally, baseline hazards (model intercepts) specific to region/ethnicity were analysed to determine if they improved probability estimates.

Prior to the multivariate model, each individual factor, namely region/ethnicity, sex, parental myopia, refractive error, axial length, AL/CR ratio and age was assessed for significance. The Harrell C statistic was used to assess the significance of each factor. The Harrell C statistic<sup>33</sup> is a measure of predictive discrimination between those that became myopic and those that remained non-myopic, using time-to-event data with censored observations, where 0.5 indicates no discrimination and 1 indicates perfect discrimination. The model assumptions, namely proportional hazards, were assessed graphically for interaction of hazard with time and using the scaled Schoenfeld residual test.<sup>34</sup> Log transformations of refractive error, axial length and AL/CR ratio were tested and retained if statistically significant and improved the model fit. To account for the correlation of eye-level data for each participant who contributed both eyes to the analysis, eye-specific data was treated as a cluster, and a robust variance-covariance estimator was computed for the Cox proportional-hazards model.<sup>35</sup> Modelling steps included forward entry of main effects

followed by backward elimination. Interactions with region/ethnicity were tested and retained in the model if statistically significant. The final model included only factors that were statistically significant. The strength of association of each factor was described using a hazard ratio (relative risk) and its 95% confidence limits. The baseline cumulative hazard for each discrete time period was natural log transformed and added to the model as the constant for 12 and 24 months, respectively. The final model equations along with the baseline cumulative hazard were then used to estimate the cumulative probability of myopia incidence at 12 and 24 months.

## Methods used to calculate hyperopic reserve

Two alternative approaches were used to estimate the level of hyperopic reserve required to minimise the risk of myopia onset. One method computed the Youden index,<sup>36</sup> which has been used previously for estimating hyperopic reserves.<sup>16</sup> For this analysis, baseline SE refraction was converted into binary variables using cut-off values of  $-0.25, 0, 0.25, 0.5, 0.75, 1, 1.25$  and  $1.5$  D, and ROC curve analysis for the 2-year myopia incidence was used to compute the Youden's index.<sup>36</sup> The Youden's index is the maximum value of sensitivity + specificity  $-1$ , or the vertical distance on the ROC curve that maximises both sensitivity and specificity when identifying a child at risk.

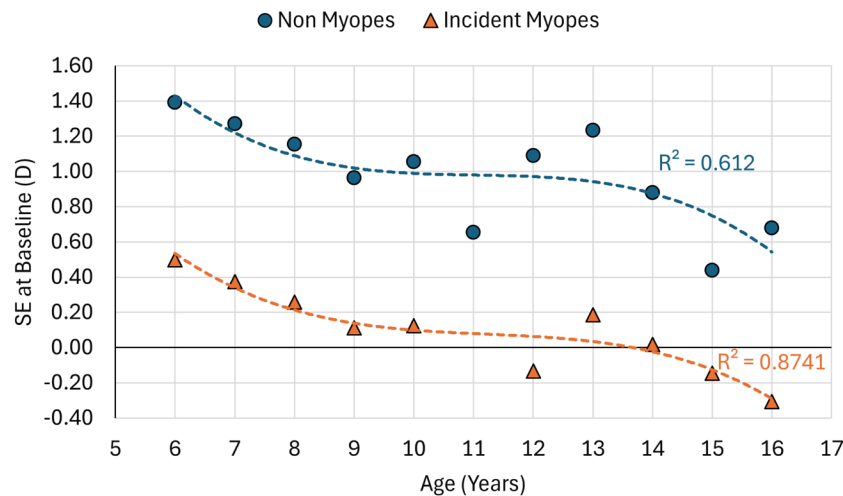
The Cox regression model was also used to identify the hyperopic reserve across age and region/ethnicity. The model equation was used to estimate the cumulative probability of myopia incidence at 24 months. For each age and region, the least hyperopic refractive error (in increments of  $0.25$  D) that had a 24-month probability of myopia incidence of  $<10\%$  was identified as the hyperopic reserve. While this cut-off was selected pragmatically in the absence of established standards in the pre-myopia literature, it aligns with common practice in other areas of health risk prediction. For example, cardiovascular risk assessment tools such as the Framingham Risk Score<sup>37,38</sup> and QRISK<sup>39</sup> use a 10-year absolute risk of  $10\%$  as a threshold for moderate risk, often prompting preventive intervention. Similarly, cancer screening<sup>40</sup> and chronic disease prevention guidelines<sup>41</sup> frequently apply similar thresholds to balance early identification with minimising unnecessary interventions.

The level of statistical significance was set at  $5\%$ . Statistical analysis was performed using IBM SPSS, v29 ([ibm.com](http://ibm.com)) and STATA, 10 ([stata.com](http://stata.com)).

## RESULTS

The dataset comprised 4405 unique non-myopic participants aged  $7.7 \pm 1.5$  years ranging from 6.0 to 16.8 years, of whom  $53\%$  were males,  $94.9\%$  Chinese and parental myopia was distributed as  $48.7\%, 33.1\%$  and  $18.2\%$  for zero, one and two myopic parents, respectively. Data were





**FIGURE 1** Mean baseline spherical equivalent (SE) (Dioptres) by age (all participants). The X-axis represents age in completed years at the time of the baseline visit. Follow-up measurements were obtained approximately 1 year later.

available for 6675 eyes, with baseline SE refraction ranging from  $-0.49$  to  $+10.0$  D (mean =  $+1.12$  D, SD =  $0.85$  D), axial length ranging from  $19.36$  to  $27.18$  mm (mean =  $22.79$  mm, SD =  $0.74$  mm) and AL/CR ranging from  $2.52$  to  $3.37$  (mean =  $2.91$ , SD =  $0.07$ ). Baseline characteristics of the participants are presented in Figure 1 and Table 2. Figure 2 shows the mean annualised change in refraction by age for incident and non-incident myopes.

## Myopia incidence and probabilities

During the analysis period, 621 eyes became myopic within 12 months, and cumulatively, 1014 eyes became myopic within 24 months. For the incident myopes, no regional/ethnic differences were observed for sex ( $p=0.40$ ), parental myopia ( $p=0.83$ ), refractive error ( $p=0.73$ ), axial length ( $p=0.95$ ), corneal curvature ( $p=0.28$ ) and AL/CR ( $p=0.09$ ), although European children were older ( $11.7 \pm 2.5$  years) than the Chinese children ( $7.7 \pm 1.3$  years,  $p<0.01$ ). However, the age ranges were between 6 and 16 years for both populations.

The univariate Kaplan–Meier probabilities of remaining non-myopic and the derived cumulative incidence at 12 and 24 months based on demographic factors and baseline ocular metrics are shown in Table 3. The probability of remaining non-myopic was 0.90 and 0.81 for Chinese children at 12 and 24 months, resulting in a cumulative incidence of 9.7% and 18.8%, respectively. For Europeans, values were 0.96 and 0.93 at 12 and 24 months, giving a cumulative incidence of 3.9% and 6.7%, respectively, with significant differences in incidence between regions/ethnicities (European < Chinese,  $p<0.001$ ). Females had a higher cumulative incidence than males by 24 months (19.9% vs. 16.6%,  $p=0.002$ ) and children with two myopic parents had a significantly higher cumulative incidence than those with one or no myopic parents (30.6%, 18.8% and 14.1% for two, one or no myopic parents, respectively,

$p<0.001$ ). Baseline age was a significant factor for myopia incidence. The cumulative incidence of myopia decreased with increasing age ( $p<0.01$ ). Table 3 shows that the cumulative incidence increased significantly when baseline refractive error was less hyperopic and there was a greater AL/CR ratio ( $p<0.001$ ).

## Cox regression time-to-event model

Table 4 shows the predictive discrimination of each factor using the C statistic of univariate time-to-event analysis of myopia incidence. All demographic variables tested were significantly associated with myopia incidence ( $p \leq 0.01$ ). For both European and Chinese children, refractive error at baseline had the greatest predictive discrimination between those that became myopic and those that remained non-myopic (C statistic 0.90,  $p<0.001$ ) followed by AL/CR ratio (C statistic 0.75,  $p<0.001$ ). Predictive discrimination was similar between ethnicities for all factors, although axial length had a slightly greater discriminatory ability in Chinese children (C statistic 0.65) compared to European children (0.55,  $p=0.04$ ).

Multivariate analysis using Cox regression, as shown in Table 5, indicates that factors associated with myopia incidence in the period up to 24 months were region/ethnicity, age at baseline, sex, parental myopia, refractive error at baseline and AL/CR at baseline, with an overall model C statistic of 0.90. No obvious violation of the model's proportional-hazards assumption was observed graphically. Also, the test for Schoenfeld residuals was non-significant for each factor (all  $p>0.10$ ). These indicate that the factor-specific relative risks were not significantly different between 12 and 24 months. The Cox regression model's factor coefficients and its associated relative risks are presented in Table 5. The factor coefficients are the linear regression coefficients for the natural log transformed hazard ( $y$ ). The following inferences are based on the multivariate model shown in



TABLE 2 Study sample and baseline characteristics by region/ethnicity.

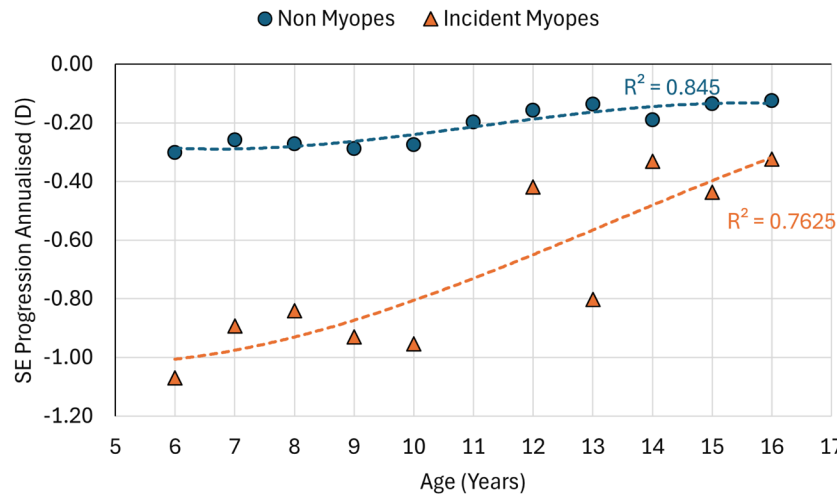
| Region/ethnicity | Study name       | Participants                         | Eyes | Age (years)                                       | Female | 2 myopic parents | Spherical equivalent (D)     | Axial length (mm)             | Corneal radius of curvature (mm) | AL/CR ratio                |
|------------------|------------------|--------------------------------------|------|---------------------------------------------------|--------|------------------|------------------------------|-------------------------------|----------------------------------|----------------------------|
| China            | Study 1 Shanghai | 2452                                 | 4513 | 7.7 ± 1.4 (6.1 to 16.8)                           | 45.1%  | 14.6%            | 1.09 ± 0.86 (−0.38 to 9.50)  | 22.77 ± 0.74 (19.36 to 27.18) | 7.83 ± 0.25 (6.80 to 8.78)       | 2.91 ± 0.08 (2.53 to 3.37) |
|                  | Study 2 Shanghai | 1730                                 | 1730 | 7.2 ± 0.6 (6.1 to 9.0)                            | 48.2%  | 22.9%            | 1.18 ± 0.73 (−0.38 to 7.00)  | 22.79 ± 0.70 (19.51 to 25.00) | 7.83 ± 0.26 (7.03 to 8.75)       | 2.91 ± 0.06 (2.52 to 3.13) |
| Europe           | Study 3 UK       | 106 (6–7 years: 53, 12–13 years: 53) | 210  | 6.9 ± 0.3 (6.2 to 7.6), 13.1 ± 0.3 (12.4 to 13.8) | 63.2%  | 11.4%            | 1.55 ± 1.34 (−0.25 to 10.00) | 22.93 ± 0.96 (19.73 to 25.19) | 7.87 ± 0.27 (7.35 to 8.43)       | 2.91 ± 0.08 (2.55 to 3.11) |
|                  | Study 4 Sweden   | 117                                  | 222  | 11.7 ± 2.5 (8.0 to 16.0)                          | 54.7%  | 21.4%            | 0.96 ± 0.98 (−0.49 to 5.69)  | 23.01 ± 0.78 (20.93 to 24.91) | 7.77 ± 0.27 (7.08 to 8.44)       | 2.96 ± 0.08 (2.67 to 3.13) |

Abbreviations: AL/CR, Axial length/Corneal radius of curvature; D, dioptres.

**Table 5.** Natural log transformed refractive error, computed as  $\ln(\text{SE} + 1)$  was a better predictor than its linear term. The risk of myopia incidence increased by 4.5 times (95% CI: 4.1 to 4.9) for every 0.5 D incremental increase of baseline refraction in the myopic direction. Greater AL/CR also increased the risk of myopia incidence, where a 0.1 incremental increase in baseline AL/CR was associated with a 1.4 times increased risk of myopia incidence (95% CI: 1.3 to 1.6). In terms of demographic factors, after adjusting for refractive error, AL/CR ratio and age, region/ethnicity had the greatest strength of association, with Chinese eyes having a higher risk than European eyes by 3.3 (95% CI: 1.87 to 5.96). Females had a 44% higher risk than males. Children with parental myopia were at a greater risk of myopia incidence, where one myopic parent conferred a 20% increased risk and two myopic parents conferred a 63% increased risk compared with children with no myopic parents. The risk of myopia incidence decreased with increasing age, where the risk decreased by 13% for every 1-year increase in baseline age. Interactions by ethnicity were analysed but were not statistically significant with age ( $p=0.94$ ), sex ( $p=0.81$ ), parental myopia ( $p=0.07$ ), refractive error ( $p=0.29$ ) and AL/CR ( $p=0.98$ ), denoting that the greater risk in Chinese children was independent of these factors.

While the full model (Table 5) provides the relative risk of each factor, further analysis indicated that baseline intercepts were greatly weighted towards the Chinese sample due to the unequal distribution by region/ethnicity. As a result, region/ethnicity specific baseline hazards were computed and observed to be 2 times higher in the Chinese sample at 12 months and 3.9 times higher at 24 months compared to the European sample. Accordingly, region specific baseline intercepts were used for the first 12-month and second 12-month periods to obtain more accurate probability estimates within each region and time period. The model coefficients with region specific baseline intercepts are provided in Table S1. Probability estimates by age and refractive error were derived from this model for 24-month myopia incidences and graphically displayed in Figure 3a–c for Chinese with zero, one or two myopic parents and Figure 3d–f for Europeans with zero, one or two myopic parents, respectively. These assume an equal sex distribution and average AL/CR of 2.91, which was the mean at baseline for the entire sample. Model-based probability estimates of myopia incidence as shown in Figure 3 indicate that the probability increases exponentially with less hyperopic baseline refractive error across all ages.

For an 8-year-old child having one myopic parent and a refractive error of 0.00 D, the probability of incident myopia by 24 months was 19% for Europeans but increased to 35% when the baseline refractive error was −0.25 D. In Chinese children, this increased from 47% to 68% between baseline refractions of 0.00 D and −0.25 D. Between 6 and 16 years of age, the probability of myopia incidence for children having one myopic parent with a 0.00 D refractive error at baseline reduced by approximately 16% (23% to 7%) for European children but 31% (54% to 23%) for Chinese children.



**FIGURE 2** Mean annualised change in spherical equivalent (SE) (Dioptres) by age (all participants). The X-axis represents age in completed years at the time of the baseline visit. Follow-up measurements were obtained approximately 1 year later.

The risk associated with parental myopia appears to increase exponentially in European children. For example, for an 8-year-old child with 0.00 D refractive error, the risk increased from 16% when there was no myopic parent to 23.8% with two myopic parents. In comparison, the increase was greater in Chinese children, ranging from 42.9% to 55.1%.

Hyperopic reserve was analysed using both the Youden index and the multivariate model. When using the Youden index, which is based on the univariate ROC curves, SE of +0.25 D had the highest Youden index (0.47) in European eyes, while +0.75 D had the highest Youden index (0.57) in Chinese eyes, indicating that a higher hyperopic reserve was required in Chinese eyes. However, since it is a univariate analysis, the confounding effect of age was not considered. For a more accurate analysis of hyperopic reserve (which adjusts for confounding factors), probability estimates from the Cox regression model were used to identify age and region/ethnicity-specific hyperopic reserve values. Figure 4 shows the hyperopic reserve for each age and region/ethnicity by parental myopia by restricting the incidence probability levels to <10% over 24 months. As observed from Figure 4, Chinese eyes required a greater hyperopic reserve than European eyes for a given age. For example, a 6-year-old Chinese child required +1.50 D compared to +0.50 D for a European child with one myopic parent, while a 16-year-old Chinese child required +0.50 D compared to 0.00 D for a European child with one myopic parent. Hyperopic reserve was dependent on age and parental myopia, where younger children and those with parental myopia required greater reserve.

## DISCUSSION

In this study, factors predictive of myopia onset in both Chinese and European children included age, parental myopia, sex, baseline SE, axial length and AL/CR. While

the factors influencing myopia incidence were similar between the two population groups, the Chinese group demonstrated a higher probability of developing myopia. This suggests that other unaccounted factors may be driving the differing probabilities of incident myopia. These differences could potentially be attributed to environmental influences or genetic factors that are not yet fully understood.

Overall, a lower SE and higher AL/CR in a non-myopic child confers a greater probability of developing myopia within 12–24 months, and the probability is greater at a younger age in the Chinese population, in females and those with parental myopia. These findings confirm previous research on predictive factors for myopia onset.<sup>16,17,19,21,22</sup>

The present study also confirms that SE is the most influential predictive factor for both European and Chinese populations.<sup>16,17,19–21</sup> Using a look-up table based on SE is the simplest practical tool for assessing a child's risk of myopia onset.<sup>23</sup> Zadnik et al.<sup>16</sup> presented data from one of the earliest published studies involving pre-myopic children in a multi-ethnic cohort in the USA, the Collaborative Longitudinal Evaluation of Ethnicity and Refractive Error (CLEERE) study. The authors used the Youden index in conjunction with the area under the ROC curve to find an optimal threshold, or cut-off point for a hyperopic reserve threshold at a given age. This method helps identify the optimal balance between sensitivity and specificity for predicting myopia onset. The hyperopic reserve cut-offs recommended by Zadnik et al.<sup>16</sup> were  $\leq +0.75$  D for 6 years of age,  $\leq +0.50$  D for 7–8 years,  $\leq +0.25$  D for 9–10 years and  $\leq 0.0$  D for 11 years of age. Studies of Asian cohorts have also presented data in relation to the level of hyperopic reserve required for protection against myopia incidence; however, it is difficult to compare with the USA data due to methodological differences in study design and analysis methods.<sup>17</sup> Some studies from Asia have suggested that their findings are similar to the CLEERE data.<sup>17,21,42</sup> However, a recent large-scale study

**TABLE 3** Univariate Kaplan–Meier estimates of probability of remaining non-myopic and cumulative incidence of myopia by baseline factors (region/ethnicity, sex, parental myopia, age, refractive error and Axial length/Corneal radius of curvature (AL/CR)).

| Factor                | Factor level     | Probability remaining non-myopic |           | Cumulative incidence |           | Log rank test, <i>p</i> value |
|-----------------------|------------------|----------------------------------|-----------|----------------------|-----------|-------------------------------|
|                       |                  | 12 months                        | 24 months | 12 months            | 24 months |                               |
| Region/Ethnicity      | Chinese          | 0.903                            | 0.812     | 9.7%                 | 18.8%     | <b>&lt;0.001</b>              |
|                       | European         | 0.961                            | 0.933     | 3.9%                 | 6.7%      |                               |
| Sex                   | Male             | 0.913                            | 0.834     | 8.7%                 | 16.6%     | <b>0.002</b>                  |
|                       | Female           | 0.900                            | 0.801     | 10.0%                | 19.9%     |                               |
| Parental Myopia       | No Myopic Parent | 0.925                            | 0.859     | 7.5%                 | 14.1%     | <b>&lt;0.001</b>              |
|                       | 1 Myopic Parent  | 0.907                            | 0.812     | 9.3%                 | 18.8%     |                               |
|                       | 2 Myopic Parents | 0.854                            | 0.694     | 14.6%                | 30.6%     |                               |
| Age (years)           | 6–7              | 0.920                            | 0.828     | 8.0%                 | 17.2%     | <b>&lt;0.001</b>              |
|                       | 8–9              | 0.870                            | 0.780     | 13.0%                | 22.0%     |                               |
|                       | 10–11            | 0.855                            | 0.840     | 14.5%                | 16.0%     |                               |
|                       | 12–13            | 0.941                            | 0.874     | 5.9%                 | 12.6%     |                               |
|                       | ≥14              | 0.899                            | 0.899     | 10.1%                | 10.1%     |                               |
| Refractive error (RE) | −0.49 to −0.01 D | 0.276                            | 0.152     | 72.4%                | 84.8%     | <b>&lt;0.001</b>              |
|                       | 0.00 to 0.49 D   | 0.651                            | 0.433     | 34.9%                | 56.7%     |                               |
|                       | 0.50 to 0.99 D   | 0.935                            | 0.784     | 6.5%                 | 21.6%     |                               |
|                       | ≥ 1.00 D         | 0.992                            | 0.953     | 0.8%                 | 4.7%      |                               |
| AL/CR                 | ≤2.7             | 0.993                            | 0.983     | 0.7%                 | 1.7%      | <b>&lt;0.001</b>              |
|                       | 2.8              | 0.989                            | 0.959     | 1.1%                 | 4.1%      |                               |
|                       | 2.9              | 0.942                            | 0.856     | 5.8%                 | 14.4%     |                               |
|                       | 3.0              | 0.816                            | 0.683     | 18.4%                | 31.7%     |                               |
|                       | ≥ 3.1            | 0.599                            | 0.494     | 40.1%                | 50.6%     |                               |

The level of significance was set at 5% (Bold Italics).

in China, which tracked children over a 16-year period, identified a hyperopic reserve threshold of +1.50 D to be associated with a reduced risk of myopia onset, regardless of age or sex.<sup>28,43</sup> They cited methodological differences in analysis methods as a factor in the contrasting findings compared to previous Asian studies and suggested that a higher hyperopic reserve threshold is appropriate for Chinese compared to White children.

There is currently a lack of consensus in relation to hyperopic reserve cut-off criteria and methodologies used to arrive at an appropriate value. Study differences include analysis methodology, follow-up period, definitions of myopia and hyperopic reserve and differences in population characteristics. The present study utilised two methodologies to explore hyperopic reserve thresholds, including that employed by Zadnik et al.<sup>16</sup> and an alternative method using probability cut-offs. When applying the Youden index as employed by Zadnik et al., the optimal cut-off in the present study was found to be +0.25 D for Europeans versus +0.75 D for Chinese. However, this is an overall average based on univariate analysis that did not account for other risk factors such as age and parental myopia. As Chinese children are more likely to develop myopia at an earlier age,<sup>44</sup> this methodology and consequent interpretation of results

are highly limited when comparing populations. Hence, the present study defines hyperopic reserve based on probabilities using a multivariate model, potentially providing a more robust and clinically meaningful application.<sup>26</sup> Using a 24-month probability cut-off of 10%, the present study showed that hyperopic reserve ranged from 0.00 D to +0.75 D in European eyes and was higher in Chinese eyes, ranging from +0.50 D to +1.50 D depending on age and risk factors such as parental myopia. Although not directly comparable, both types of analysis support the finding that a higher hyperopic reserve is required in Chinese compared to European children. To enable meaningful comparisons across studies and for guiding early management strategies for myopia prevention, wider consensus on methodological approaches is needed in the field of pre-myopia. It is acknowledged that risk thresholds may vary depending on the purpose of the analysis and the population for which it is used. In recent years, probability models like the one outlined in the present paper have been the basis for the development of more sophisticated online 'calculators' for profiling a child at risk of myopia based on multiple risk factors rather than relying on SE alone. These are summarised in Table 6 and compared with the approach taken in the present study. All calculators use



**TABLE 4** Harrell's C statistic (Area under the Receiver Operating Characteristic (ROC) curve) for univariate survival analysis of incident myopia.

| Factor at baseline          | Harrell's C statistic |                  | C statistic between region/ethnicity |       |                |
|-----------------------------|-----------------------|------------------|--------------------------------------|-------|----------------|
|                             | Full sample           | <i>p</i> value   | Europe                               | China | <i>p</i> value |
| Refractive error            | 0.90                  | <b>&lt;0.001</b> | 0.86                                 | 0.90  | 0.50           |
| AL/CR ratio                 | 0.75                  | <b>&lt;0.001</b> | 0.76                                 | 0.75  | 0.61           |
| Axial length                | 0.63                  | <b>&lt;0.001</b> | 0.55                                 | 0.65  | <b>0.04</b>    |
| Parental myopia             | 0.60                  | <b>0.001</b>     | 0.60                                 | 0.62  | 0.06           |
| Region/ethnicity            | 0.58                  | <b>0.001</b>     |                                      |       |                |
| Corneal radius of curvature | 0.57                  | <b>&lt;0.001</b> | 0.69                                 | 0.59  | 0.30           |
| Age                         | 0.55                  | <b>0.01</b>      | 0.55                                 | 0.57  | 0.32           |
| Sex                         | 0.54                  | <b>0.01</b>      | 0.58                                 | 0.58  | 0.92           |

Abbreviation: AL/CR: Axial length/Corneal radius of curvature.

The level of significance was set at 5% (Bold Italics).

**TABLE 5** Cox regression model equation and relative risks of factors associated with myopia incidence over 2 years. Model equation includes intercepts specific for 12 and 24-month discrete periods.

| Factor                                                             | Coefficient <sup>a</sup> | <i>p</i> -value  | Hazard ratio (relative risk) | 95.0% confidence interval |       |
|--------------------------------------------------------------------|--------------------------|------------------|------------------------------|---------------------------|-------|
|                                                                    |                          |                  |                              | Lower                     | Upper |
| Model intercept for 12 months                                      | -11.68                   |                  |                              |                           |       |
| Model intercept for 24 months                                      | -10.58                   |                  |                              |                           |       |
| Age at baseline (every 1-year increase)                            | -0.139                   | <b>&lt;0.001</b> | 0.870                        | 0.823                     | 0.920 |
| Male                                                               | 0.000                    |                  | 1 (Reference)                |                           |       |
| Female                                                             | 0.363                    | <b>&lt;0.001</b> | 1.438                        | 1.261                     | 1.640 |
| Region/Ethnicity: European                                         |                          |                  |                              |                           |       |
| Region/Ethnicity: Chinese                                          | 1.205                    | <b>&lt;0.001</b> | 3.336                        | 1.868                     | 5.959 |
| Parental Myopia: None                                              | 0.000                    |                  | 1 (Reference)                |                           |       |
| Parental Myopia: One parent                                        | 0.181                    | <b>0.02</b>      | 1.198                        | 1.026                     | 1.400 |
| Parental Myopia: Two parents                                       | 0.486                    | <b>&lt;0.001</b> | 1.625                        | 1.383                     | 1.909 |
| Refractive Error at baseline: for every 1 D increase of LN(SE + 1) | -3.000                   | <b>&lt;0.001</b> | 0.050                        | 0.041                     | 0.060 |
| AL/CR ratio at baseline (every 0.1 increase in AL/CR)              | 0.344                    | <b>&lt;0.001</b> | 1.411                        | 1.267                     | 1.571 |

Abbreviations: AL/CR, Axial length/Corneal radius of curvature; D, Dioptres; LN, Natural log; SE, Spherical equivalent.

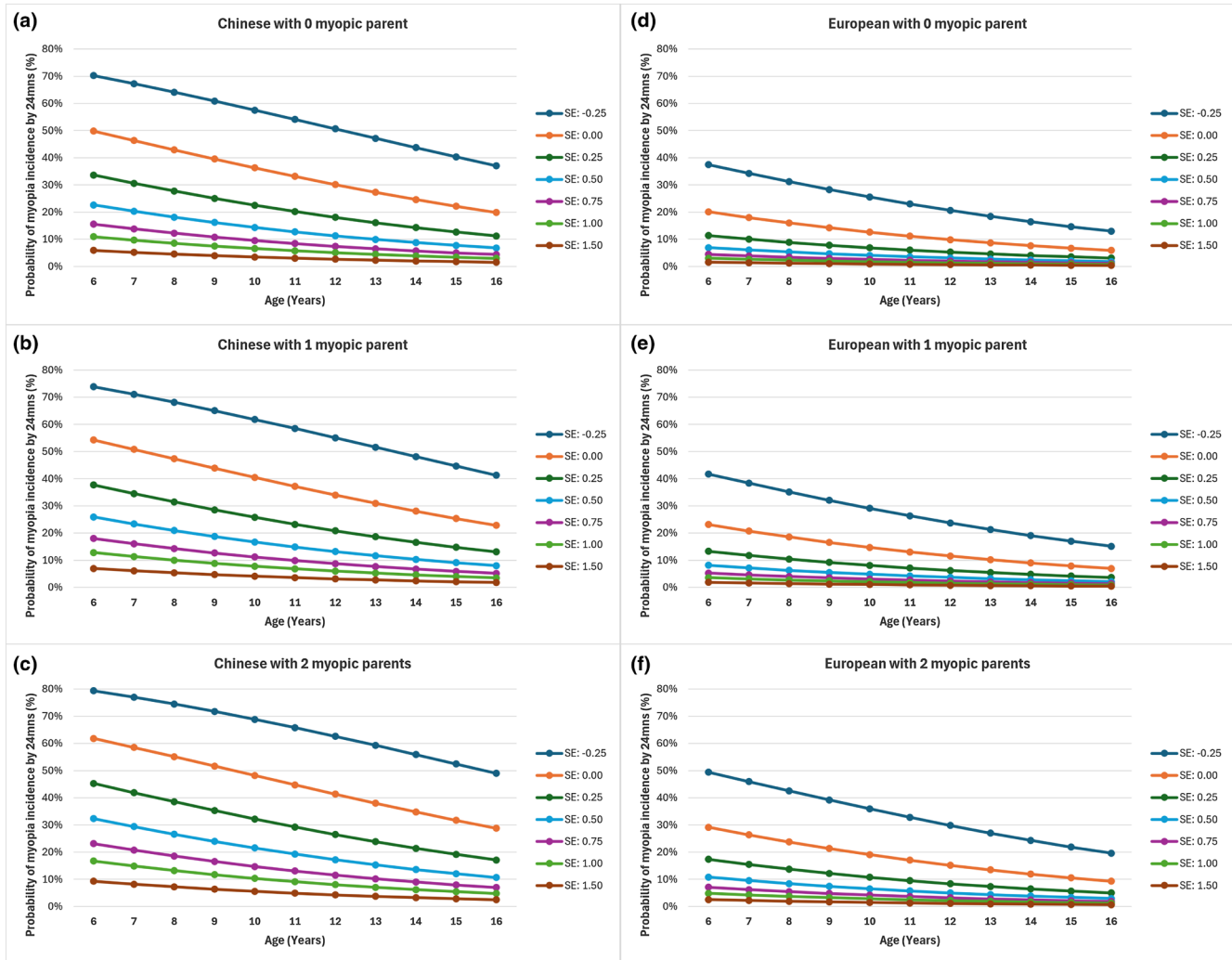
The level of significance was set at 5% (Bold Italics).

<sup>a</sup>Linear regression coefficients to predict natural log of hazard.

SE as a basis for predictions, but the inclusion of other predictive and/or risk factors varies.

Using SE alone as a guide to predict myopia would be helpful in practice, particularly since many clinicians do not have access to optical biometers to assess axial length.<sup>45</sup> Whilst refractive error alone is sufficient to discriminate adequately between incident myopes and non-myopes, AL/CR is an additional significant factor and acts as a confounder to SE in this multi-ethnic dataset, and thus helps to refine individual prediction probabilities by capturing aspects of ocular biometry related to eye growth that are not fully reflected by refractive error alone. Moreover, the current study found that AL/CR had

a high C statistic (0.75) and contributed significantly to the multivariate model ( $p < 0.001$ ). These refinements in probability estimates offer additional clinical utility, enabling clinicians to use a risk calculator that generates individualised risk scores rather than relying solely on SE measurements. However, axial length (hazard ratio 1.02,  $p = 0.78$ ) and corneal radius of curvature (hazard ratio 1.04,  $p = 0.82$ ) did not improve the model significantly. In situations where AL/CR is not readily available, the sample average value of 2.91 could be used due to the confounding nature of AL/CR. Although baseline AL/CR was not significantly different between ethnicities when adjusting for age and refractive error, verification of this



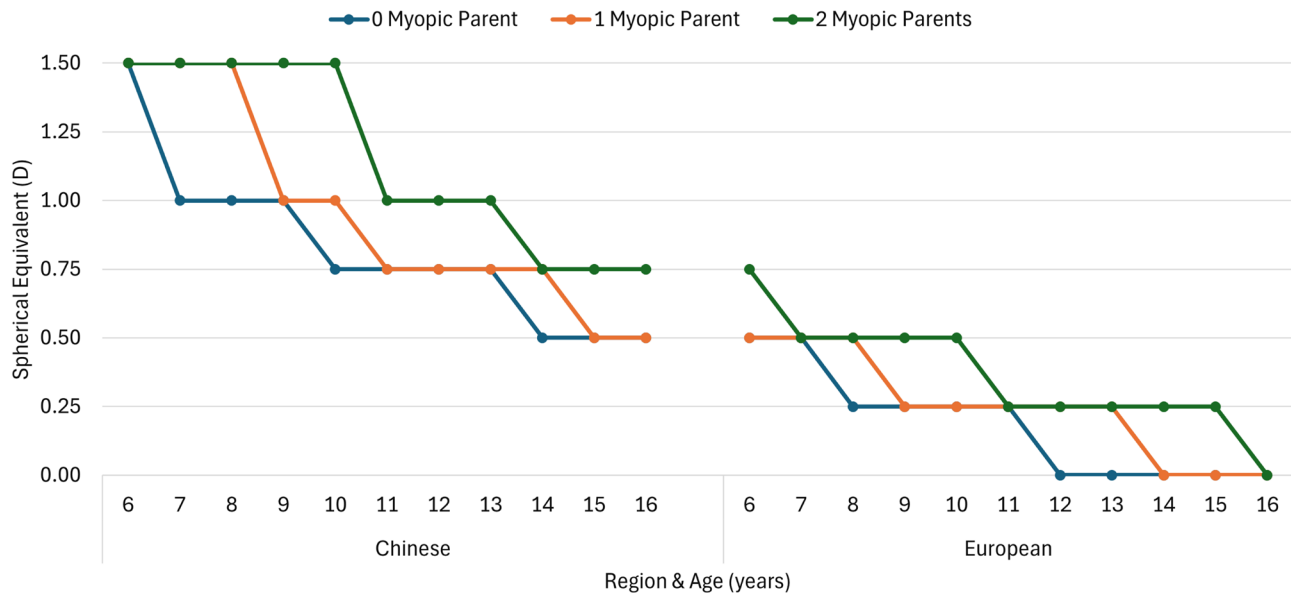
**FIGURE 3** Cox regression probability of myopia incidence by 24 months from baseline by age and refractive error assuming equal sex distribution and average axial length / corneal radius ratio (AL/CR). (a–c) East-Asian eyes with 0, 1 or 2 myopic parents, respectively. (d–f) European eyes with 0, 1 or 2 myopic parents, respectively. The X-axis represents age in completed years at the time of the baseline visit. Follow-up measurements were obtained approximately 1 year later.

suggestion is required. Mutti et al.<sup>26</sup> have developed a myopia incidence predictive probability model based on the CLEERE study. In their model, they excluded axial length and corneal power, reporting that these did not add significantly to predictions based on refractive error alone. However, the PreMO<sup>27</sup> tool's predictive ability was enhanced when axial length and parental myopia were added, compared to SE alone. Differences in model factors may reflect population variations and differences in study methodology (e.g., CLEERE used ultrasound biometry). Hence, the availability and use of population-specific tools for myopia incidence prediction and protocol standardisation are warranted.

From the available online calculators, the only other predictive tool that incorporates ethnicity as an input variable is the CLEERE calculator. However, as previously mentioned, these data are from ethnically diverse populations living in the USA, whereas the data from the present

study report on different ethnic populations living in their respective country of origin, that is, Chinese living in China versus Europeans living in Europe. According to the literature, Asians living in Asia may have different environmental and hence different probabilities of incident myopia than those living elsewhere.<sup>25</sup> Hence, this study provides additional nuance in respect to regional/ethnic differences.

We recognise several limitations to the present study. Firstly, the present study had a limited follow-up period of 24 months, and it is likely that some children categorised into the non-incident myope group will develop myopia in later years. Using the area under the ROC to indicate accuracy, Mutti et al.<sup>26</sup> found model predictions to be most accurate in the first year, with a reduction in subsequent years. The current model also found a higher area under the ROC for year 1 than year 2 (0.93 vs. 0.89). It could be argued that the most critical period for prevention is the year



**FIGURE 4** Hyperopic reserve by age, parental myopia and region/ethnicity. The X-axis represents age in completed years at the time of the baseline visit. Follow-up measurements were obtained approximately 1 year later.

immediately prior to onset, as this has been found to be the period that the eye undergoes the most rapid growth.<sup>8</sup> Short-term predictions are important for clinical decision making, particularly in respect to prescribed follow-up periods which may be shorter for a higher risk patient than one deemed to be at lower risk.

A second limitation of this study is the absence of 24-month follow-up data from the UK cohort. The 12-month incidence of myopia in the UK sample (2.9%) was slightly lower than in the Swedish sample (4.9%), but the difference was not statistically significant ( $p = 0.26$ ). The combined 12-month incidence for both samples was 3.91%. In the Swedish cohort, as expected, the incidence during the second 12-month period was lower than during the first (2.9% vs. 4.9%). Given this trend, the second-year incidence in the UK is also likely to be low. Therefore, the absence of 24-month data from the UK cohort is unlikely to affect the combined incidence estimates significantly. However, caution is needed when interpreting the 24-month probability estimates for Europe due to the smaller sample size overall. Additional data from other European regions would improve the reliability and generalisability of these estimates.

The use of the Cox proportional-hazards model for this analysis is a third potential limitation due to the short duration and use of discrete time intervals, as the Cox model assumes that time-to-event data are continuous. In studies with discrete time periods, alternative models, such as the grouped proportional-hazards model with a complementary log-log link, may be more appropriate.<sup>46</sup> However, the Cox model was chosen here because the proportion of tied incidence times was low (<20%), consistent with recommendations by Chalita et al.<sup>47</sup> Tied incidence times occur when multiple participants

experience the event (e.g., onset of myopia) at the same recorded time, such as at the end of a 6- or 12-month follow-up period, rather than at distinct, continuous time points. In scenarios with many tied times, the Cox model may perform less effectively due to its assumption of continuous time. However, when the proportion of tied times is relatively low, as in this study, then the Cox model remains a valid and robust choice for analysing time-to-event data.

A fourth limitation of the present study is the lower number of incident myopes in the European population which also had an older age distribution than the Chinese population. However, when adjusting for age, the present model revealed no significant difference in the influence of risk factors and their level of predictive importance between the two regions. Thus, the impact of the size of the European sample may be low, although additional European studies will be required to confirm these findings.

A fifth study limitation is that other predictive metrics that have been identified in the literature were not included in this study. These include crystalline lens power,<sup>48</sup> outdoor time and near work,<sup>20</sup> accommodation,<sup>49</sup> astigmatism,<sup>26</sup> body height and weight.<sup>42</sup> However, the impact of the addition of these variables is likely to be small based on previous research findings and the fact that the model presented has a high level of accuracy (C statistic of 0.9). A final limitation of the study is the potential for sampling bias. Although population-based strategies were employed, differences between sites – such as the need for travel in Sweden – may have led to greater participation from families with heightened concern about vision. Nonetheless, the predictive model was developed to account for key risk factors, such as parental myopia, which helps mitigate the impact of such bias.



TABLE 6 List of calculators to predict onset of myopia compared for model variables and features.

| Calculators                                                                                                           | C statistic/area under ROC curve | Age | Sex | Parental myopia | Ethnicity | SE  | AL  | CR  | AL/CR ratio | Other         | Yearly prediction increments | Maximum prediction duration | Able to compute probability? |
|-----------------------------------------------------------------------------------------------------------------------|----------------------------------|-----|-----|-----------------|-----------|-----|-----|-----|-------------|---------------|------------------------------|-----------------------------|------------------------------|
| Wenzhou Medical University Essilor Progression and Onset of Myopia (WEPrOM)                                           | 0.74                             |     |     |                 |           | Yes | Yes | Yes |             | Accommodation | 1 year                       | 3.5 years                   | Yes                          |
| Northern Ireland Childhood Errors of Refraction (NICER) study and its Predicting Myopia Onset and progression (PreMO) |                                  | Yes |     | Yes             |           | Yes | Yes |     |             |               | 3 years                      | 10 years                    | No (Scores)                  |
| Singapore Cohort Study of the Risk factors for Myopia (SCORM)                                                         | 0.91                             | Yes |     | Yes             |           | Yes |     |     |             |               | 1 year                       | 1 year                      | No (Scores)                  |
| Collaborative Longitudinal Evaluation of Ethnicity and Refractive Error (CLEERE)                                      | 0.83–0.92                        | Yes | Yes |                 | Yes       | Yes |     |     |             | J0            | 1 year                       | 8 years                     | Yes                          |
| Current publication                                                                                                   | 0.90                             | Yes | Yes | Yes             | Yes       | Yes |     |     | Yes         |               | 1 year                       | 2 years                     | Yes                          |

Note: 'Yes' refers to model variables that were significantly associated with myopia incidence. Empty cells refer to variables that were either not significant or not analysed.

Abbreviations: AL, axial length; CR, corneal radius of curvature; ROC, receiver operating characteristic; SE, spherical equivalent.



## CONCLUSIONS

The regional risk of myopia onset can be modelled based on age, sex, parental myopia, baseline SE and AL/CR for both Asian and European children. The influence of predictive risk factors on incident myopia was similar between Chinese and Europeans. While the factors influencing myopia incidence were similar between the two populations, the Chinese group demonstrated a higher probability of developing myopia. This suggests that other unaccounted factors may be driving the differing probabilities of incident myopia. These differences could potentially be attributed to environmental influences or genetic factors that are not yet fully understood.

Of the factors investigated, baseline SE is the strongest predictor of myopia onset within a 12–24-month period for both the European and Chinese populations. This supports findings from previous research.<sup>16,17,19–21</sup> This study also used two methodologies to explore hyperopic reserve thresholds that may be protective against myopia incidence. Both methods provide evidence that a higher hyperopic reserve is required in Chinese children compared to European children. The hyperopic reserve requirement also varies based on age and other risk factors for both population groups.

The data from the present study could be used to derive region/ethnicity specific predictive probabilities of a child developing myopia within a 12–24 month period.

## AUTHOR CONTRIBUTIONS

**Nina Tahhan:** Conceptualization (lead); data curation (supporting); formal analysis (supporting); funding acquisition (equal); investigation (equal); methodology (equal); project administration (lead); supervision (lead); writing – original draft (lead); writing – review and editing (lead). **Xiangui He:** Data curation (equal); funding acquisition (equal); investigation (equal); methodology (supporting); project administration (supporting); resources (equal); writing – review and editing (supporting). **Kathryn Saunders:** Data curation (equal); funding acquisition (equal); investigation (equal); methodology (supporting); project administration (supporting); resources (equal); writing – review and editing (supporting). **Pelsin Demir:** Data curation (equal); funding acquisition (equal); methodology (supporting); project administration (supporting); resources (equal); writing – review and editing (supporting). **Rebecca Leighton:** Data curation (equal); investigation (equal); methodology (supporting); resources (supporting); writing – review and editing (supporting). **Sara McCullough:** Data curation (equal); investigation (equal); methodology (supporting); resources (supporting); writing – review and editing (supporting). **Karthikeyan Baskaran:** Data curation (equal); funding acquisition (equal); methodology (supporting); project administration (supporting); resources (equal); writing – review and editing (supporting). **Antonio Filipe Macedo:** Data curation (equal); funding acquisition (equal); methodology (supporting); project administration

(supporting); resources (equal); writing – review and editing (supporting). **Xun Xu:** Data curation (equal); funding acquisition (equal); investigation (equal); methodology (supporting); resources (equal). **Padmaja Sankaridurg:** Data curation (supporting); formal analysis (supporting); methodology (supporting); supervision (supporting); writing – review and editing (supporting). **Thomas Naduvilath:** Conceptualization (supporting); data curation (lead); formal analysis (lead); funding acquisition (equal); investigation (equal); methodology (equal); project administration (equal); resources (equal); software (equal); supervision (equal); writing – original draft (supporting); writing – review and editing (equal).

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## CONFLICT OF INTEREST STATEMENT

Thomas Naduvilath: Employee of Brien Holden Vision Institute (BHVI); Nina Tahhan: Consultant at BHVI; Xiangui He and Xun Xu: Employees of Shanghai Eye Disease Prevention and Treatment Centre (SEDPTC). BHVI has a collaboration agreement with SEDPTC. BHVI hosts the BHVI myopia calculator, a free web-based myopia tool and has commercial interests in myopia management. Kathryn Saunders and Sara McCullough: Employees of Ulster University, host PreMO app, a free web-based myopia tool and receive industry funding for myopia research. Padmaja Sankaridurg: Former employee of BHVI, current employee of ZEISS VisionCare, a commercial entity involved in myopia management. No other conflict of interest relevant to this manuscript exists for any of the authors.



## ORCID

Nina Tahhan <https://orcid.org/0000-0002-8856-2483>

Xiangui He <https://orcid.org/0000-0002-8938-1879>

Kathryn Saunders <https://orcid.org/0000-0002-9289-5731>

Pelsin Demir <https://orcid.org/0000-0002-9970-3237>

Rebecca Leighton <https://orcid.org/0000-0002-5562-9055>

Sara McCullough <https://orcid.org/0000-0002-4438-1154>

Karthikeyan Baskaran <https://orcid.org/0000-0002-3745-0035>

Antonio Filipe Macedo <https://orcid.org/0000-0003-3436-2010>

Xun Xu <https://orcid.org/0000-0002-4246-4343>

Thomas Naduvilath <https://orcid.org/0000-0001-8813-2048>

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## SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

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